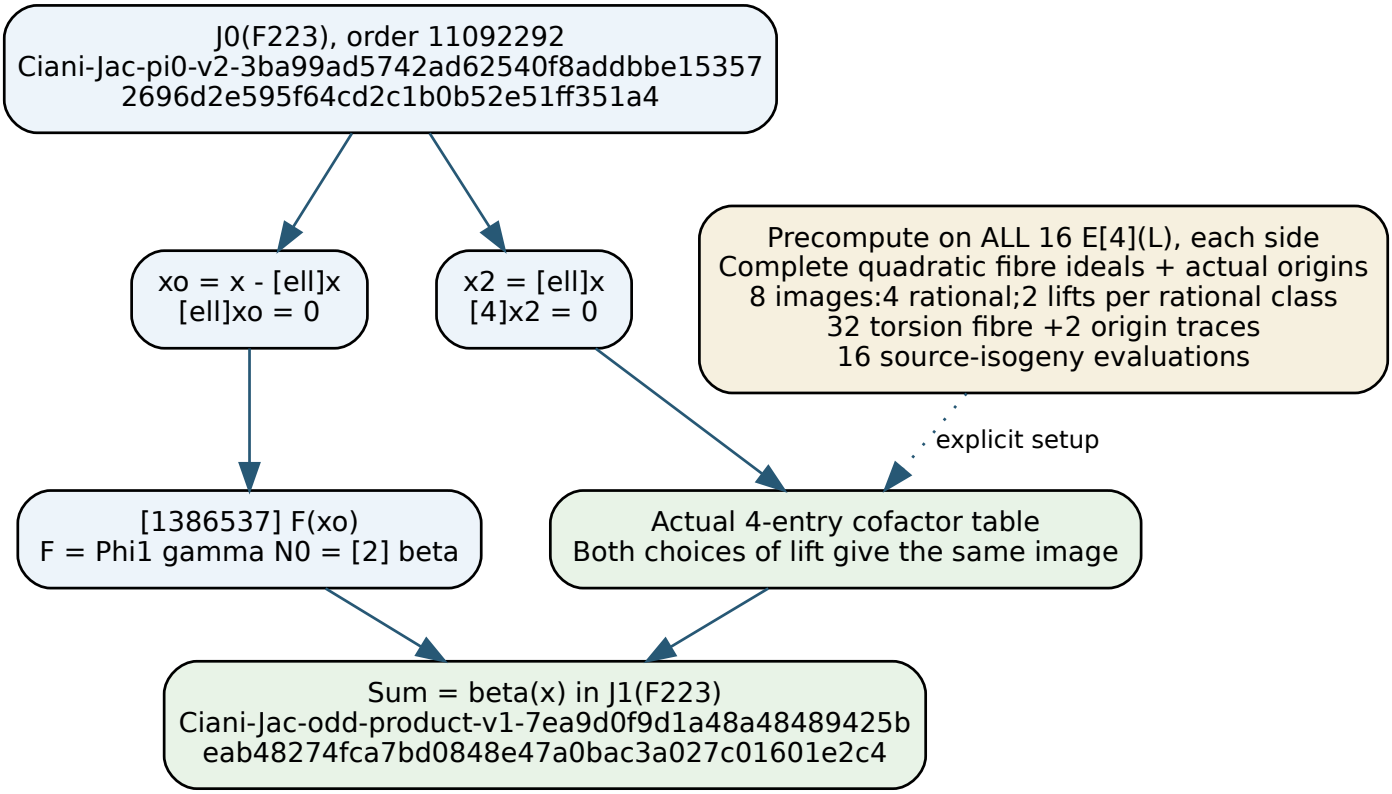
Complete rational degree-13 Jacobian transfer

2026-11-14 · native 3f1ade · conditional Lean 1e2420 · unsigned



K=F223[z]/(z³−3); E: a=(214,144,205), b=(137,218,9). E13: a=(196,168,69), b=(8,82,207).

C0, original monomial order: [0, 1, 91, 68, 209, 30, 216, 142, 31, 72, 184, 179, 36, 124, 140]; P0=[1, 0, 0].

C1, same order: [1, 121, 174, 193, 172, 175, 11, 124, 77, 78, 48, 58, 89, 63, 164]; P0=[1, 0, 84].

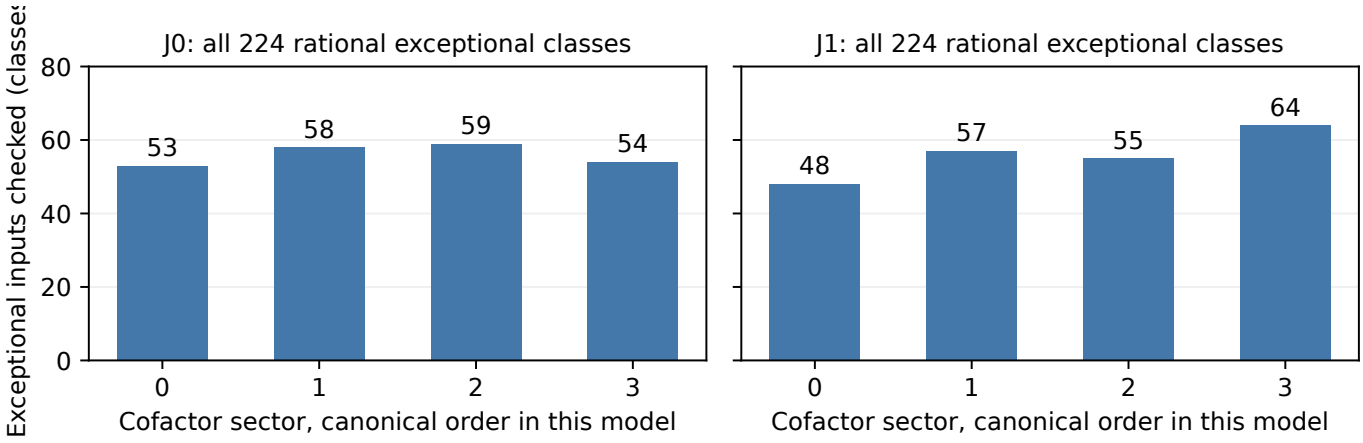
Implemented: a reusable full-rational evaluator for the matched HLP(1,0) quotient, including the cofactor-four classes. Native 3f1ade passes three unfiltered tests with zero failed/ignored/filtered and845 original guards unchanged. The full-tree audit checks exactly three accounting metadata changes; every other class, scalar and control field equals the prior records. Old raw/golden data is retained; the corrected whole fixture has no masks. The diagram shows the implemented split; the written global quotient has degree13³ and both dual compositions[13].

For ell=2773073≡1 modulo4, split x2=[ell]x and xo=x−x2. Then4x2=0 andell·xo=0. Evaluate the odd component as [1386537]F(xo), and the even component through the actual four-entry lift table. Their sum is beta(x). The full rational inverse is [3413013]betaDual; source recovery on the odd subgroup uses319970 as the inverse of26.

Quadratic preprocessing uses L=F223[w]/(w⁶−3), K=F223[z]/(z³−3), z=w²; F223² is span(1,w³). All three F²-conjugate line/conic ideals, multiplicities, infinity branches and actual origin are retained. Coefficient traces use1,w²,w⁴, and reduction uses genuine quadratic Picard arithmetic. All16 four-torsion inputs each side,256 addition pairs each, both kappa branches, original K agreement and F³ equivariance pass.

Only four of the eight quadratic images are F223-rational. In the group proof, the source group is the preimage of rational J0 inside A(F223²), not all quadratic source points. Each rational cofactor class has two actual lifts; their matched gamma/delta images agree. No scalar witness, fitted translation, reduced-support substitution or odd-DLP table enters the runtime.

Explicit input coverage and proof scope



Native coverage: 48 mixed-sector classes (six fixed odd scalars × four sectors × two directions), 96 ordered addition pairs, all five literal public targets × four sectors, and all 448 rational exceptional inputs. Both [13] compositions, the full inverse and F=[2]beta/G=[2]betaDual pass. Exceptional outputs number 1 per model. Sector indices are ordered independently by each model's canonical keys; the chart is exhaustive coverage, with no sampling uncertainty and no timing data.

Faithful conditional algebra1e2420: ordinary group/homomorphism laws, natural multiplication defined from addition, rational decomposition, odd normalization, evaluate=beta for every class, full inverse13 and DLP equivalence in both directions. Both original Lean 4.34.1/4.35.0 gates accept 44 declarations with zero trusted; all 19 named roots use no axioms beyond propext. Actual geometric/order/finite-lift/reducer bridges remain explicit written/native hypotheses.

The scalar projection $e^2-e=4me$, $e=4m+1$, has separate exact Int proof and original native certificate IDC1h2576687cb0806346: exact integer coefficient expansion, 64 sealed plus32 extra checker points, issue/replay0 and actual missing-term rejection101. Both original gates accept one Int declaration/zero trusted. The checker itself does not execute Lean. Six group bindings plus 4036be bring the partial unique total to 73; none is signed main standing.

Preserved negatives: the old setup label counted the full verification panel instead of the filtered constructor. Dynamic counts3f1ade and an actual legacy32 rejection101 correct this; the strict old fixture still rejects101. Initial Lean5f5a15 and CRT callera284e5 also remain preserved. The whole-record comparator rejects101 when the entire exceptional record is omitted, without writing an output; all 3 records are required. Earlier path, workspace, field-ring, wrong tentative DLP and same-product chain-collapse results remain intact.

Selected setup calls are 32 torsion fibre evaluations, two separate quadratic origin traces and 16 torsion source-isogeny evaluations. Both four-entry tables, all original/context setup and validation remain in the implementation and unpriced. These are selected entry counts, not a primitive-cost ledger. Calibrated cold/online costs and computational advantage are not established. This report covers the complete rational-point evaluator; global closed rational coordinates and actual geometric formalization remain separate. Next: dedicated replay/fresh full/own CI, whole genuine custody, then a prospectively reviewed optimization or genuine next chained map.

Sources: BETA_COMPLETE_RATIONAL_DERIVATION_02; BETA_COMPLETE_RATIONAL_PROOF_MAP_01; evidence/beta-complete-accounting-acceptance-05 (whole native and JSON); evidence/beta-complete-Lean-acceptance-03; beta-complete-rational-bindings-01; beta-CRT-projection-acceptance-02. Earlier 19-workflow CI cd2e03 applies to c77f95dc; this evaluator's dedicated CI is pending.